

Targeting the tumour vasculature: from vessel destruction to promotion

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Abstract

As angiogenesis was recognized as a core hallmark of cancer growth and survival, several strategies have been implemented to target the tumour vasculature. Yet to date, attempts have rarely been so diverse, ranging from vessel growth inhibition and destruction to vessel normalization, reprogramming and vessel growth promotion. Some of these strategies, combined with standard of care, have translated into improved cancer therapies, but their successes are constrained to certain cancer types. This Review provides an overview of these vascular targeting approaches and puts them into context based on our subsequent improved understanding of the tumour vasculature as an integral part of the tumour microenvironment with which it is functionally interlinked. This new knowledge has already led to dual targeting of the vascular and immune cell compartments and sets the scene for future investigations of possible alternative approaches that consider the vascular link with other tumour microenvironment components for improved cancer therapy.

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Introduction

The vasculature transports blood and oxygen and removes waste and carbon dioxide from healthy and diseased organs¹. In addition to the conduit functions of blood vessels, endothelial cells (ECs) that line blood vessels and supporting pericytes and other mural cells perform multiple tasks, including regulation of blood flow, vascular homeostasis, leukocyte and hormone trafficking, inflammation and haemostasis^{2–4}. Tumour ECs (TECs) produce angiocrine cues and tumour pericytes produce pericrine cues, which can be both paracrine and juxtacrine, and play critical roles in vascular diseases, including impacting cancer progression and therapy resistance^{5–12}. **Tumour vascularization commonly involves both tumour cell co-option of the vasculature (Box 1) and hypoxia-induced angiogenesis**, commonly involving EC proliferation, migration, sprout formation and extension to luminate vessels that are ultimately decorated with pericytes and mural cells in the process of neoangiogenesis^{13–16}. Indeed, angiogenesis is a hallmark of cancer, occurring in a tumour stage-dependent and type-dependent manner^{17–19}. Cancer blood vessels are typically tortuous, with poorly organized pericyte coverage, hyperdilated and leaky with aberrant blood flow hindering drug and oxygen delivery while supporting

tumour cell metastasis^{20–22}. In addition, interstitial pressure increases, and the resulting acidic, hypoxic tumour augments a pro-angiogenic feedback loop likening tumours to ‘wounds that never heal’^{23,24}. Since the early description of tumour blood vessel functions²⁰, we now know that ECs and their mural counterparts modulate additional functions in tumours^{5,25,26}, which are partly based on their organ-specific properties^{27–29}. Consequently, therapeutic targeting of the tumour vasculature impacts various aspects of the tumour phenotype because it is a multifaceted key component that is intimately and functionally connected to the entire heterogeneous tumour ecosystem.

This Review will not describe in detail the plethora of vascular targeting agents and drugs that have had translational impact as well as their applications and limitations. Instead, it will position anti-angiogenic therapies in the context of alternative vascular modulation concepts that have arisen based on the more recently expanded understanding of the field of tumour blood vessel functions. Our aim is to introduce the reader to, and consider, these new ideas, most of which have not been tested yet in the clinical setting, but which could provide the foundations for alternative vascular targeting strategies to improve cancer therapies.

Box 1 | Vessel co-option: a non-angiogenic mechanism of tumour vascularization

Vessel co-option refers to a phenomenon in primary tumours and metastases in which cancer cells utilize pre-existing blood vessels to obtain nutrients and oxygen for their growth and spread³²⁸. Instead of promoting angiogenesis, cancer cells become motile and migrate along normal blood vessels within the surrounding tissue of the organ. Some tumours also exploit co-option as an adaptation mechanism to become less susceptible and even resistant to conventional and anti-angiogenic treatments^{63,64,89,329}. Vessel co-option has been observed in various types of cancers, including primary lung, brain and liver cancers and metastases^{330,331}. How tumour cells enable perivascular growth is not fully understood, but cancer cells need to adhere to the basement membrane of endothelial cells or pericytes for perivascular invasion^{332–335}. A commonality appears to be the induction of an epithelial-to-mesenchymal transition program that facilitates cell migration and involves changes in cancer cell motility, in the composition of the extracellular matrix and the crosstalk with multiple adhesion molecules such as integrins^{336–340}. Recent transcriptomics data in lung tumours have shown that co-opted tumour vessels are not affected by tumour cells as they do not proliferate nor undergo morphological remodelling, and they transcriptionally resemble their counterparts in the normal organ³⁴¹. Thus, it is not surprising that vessel-co-opting malignant lesions in the lung, dermis, liver and kidney were not susceptible to anti-angiogenic therapy, which targets proliferating endothelial cells^{342–348}. Interestingly, however, remodelling of the immune microenvironment occurs in vessel co-option-dependent tumours, in which increased subsets of pro-tumoural macrophages³⁴¹ and neutrophils³⁴⁹ have been observed. Albeit challenging, finding ways to disrupt the interaction among cancer cells, blood vessels and perhaps immune cells in vessel co-opting tumours will provide more promising avenues for cancer treatment.

Anti-angiogenic therapy

Over 55 years ago, the late Judah Folkman’s notion to ‘starve a tumour to death’ by eliminating the tumour vasculature led to the conception of anti-angiogenic therapy³⁰. One of the major advantages of this idea included its direct effects on ‘normal’ diploid vascular cells, bypassing resistance to radiotherapy and chemotherapy in malignant cells and thus providing efficacy in tumours, which had already become resistant to standard therapies. With this specificity in mind, anti-angiogenic therapy should also not provoke the substantial toxic side effects that are commonly seen with many chemotherapeutic drugs targeting proliferating cells in general^{31,32}. The successes and limitations of drugs targeting major pro-angiogenic factors such as vascular endothelial growth factor (VEGF), angiopoietin, platelet-derived growth factor (PDGF), fibroblast growth factor (FGF) and MET have already been described in multiple seminal reviews^{33–35} (Fig. 1a). Several of the small molecule inhibitors or monoclonal antibodies to pro-angiogenic receptor tyrosine kinases showed efficacy in clinical trials and were subsequently approved¹⁶. The recombinant humanized monoclonal antibody bevacizumab (also known as avastin) targeting VEGFA was the first angiogenic inhibitor to be FDA-approved in 2004 (manufactured by Genentech, Inc.)^{36,37} and showed clinical activity in combination with cytotoxic chemotherapy in metastatic colorectal cancer^{38,39}, non-small-cell lung cancer (NSCLC)⁴⁰ and breast cancer^{41,42}. As a single agent, bevacizumab demonstrated activity in metastatic renal cell carcinoma⁴³ and glioblastoma^{44–46}, and in 2017, after the patent of bevacizumab expired, the FDA approved the first US biosimilar MVASI from Amgen, Inc. for the treatment of multiple cancer types⁴⁷.

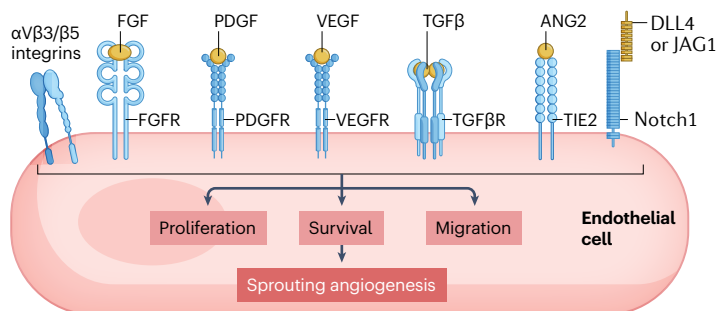
Exploiting the identification of endogenous anti-angiogenic molecules (such as thrombospondin-1 (ref. 48) and circulating protein fragments of plasminogen, extracellular matrix⁴⁹ or vascular basement membrane proteins including angiostatin, endostatin and tumstatin⁵⁰) spurred the development of a new wave of anti-angiogenic strategies with efficacy in preclinical models. Endostatin failed in US-based clinical trials⁵¹ but Endostar, a modified recombinant form of human endostatin, was approved in China in combination with chemotherapy for the treatment of patients with NSCLC^{52,53}.

In two ground-breaking papers, low-dose and incessant metronomic chemotherapy exerted antitumour effects indirectly through

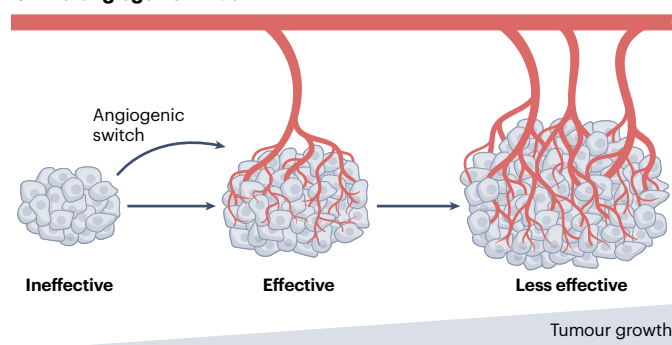
blocking angiogenesis without eliciting the toxic side effects observed with conventional maximum tolerated dose (MTD) regimens. Browder et al.⁵⁴ demonstrated anti-angiogenic effects with recurrent,

low-dose cyclophosphamide treatment in mice bearing Lewis lung carcinomas and epithelial-to-mesenchymal transition (EMT)-6 breast carcinomas. Similarly, Klement et al.⁵⁵ used metronomic dosing of the

a Anti-angiogenic targets



b Anti-angiogenic window



c Anti-angiogenic effects

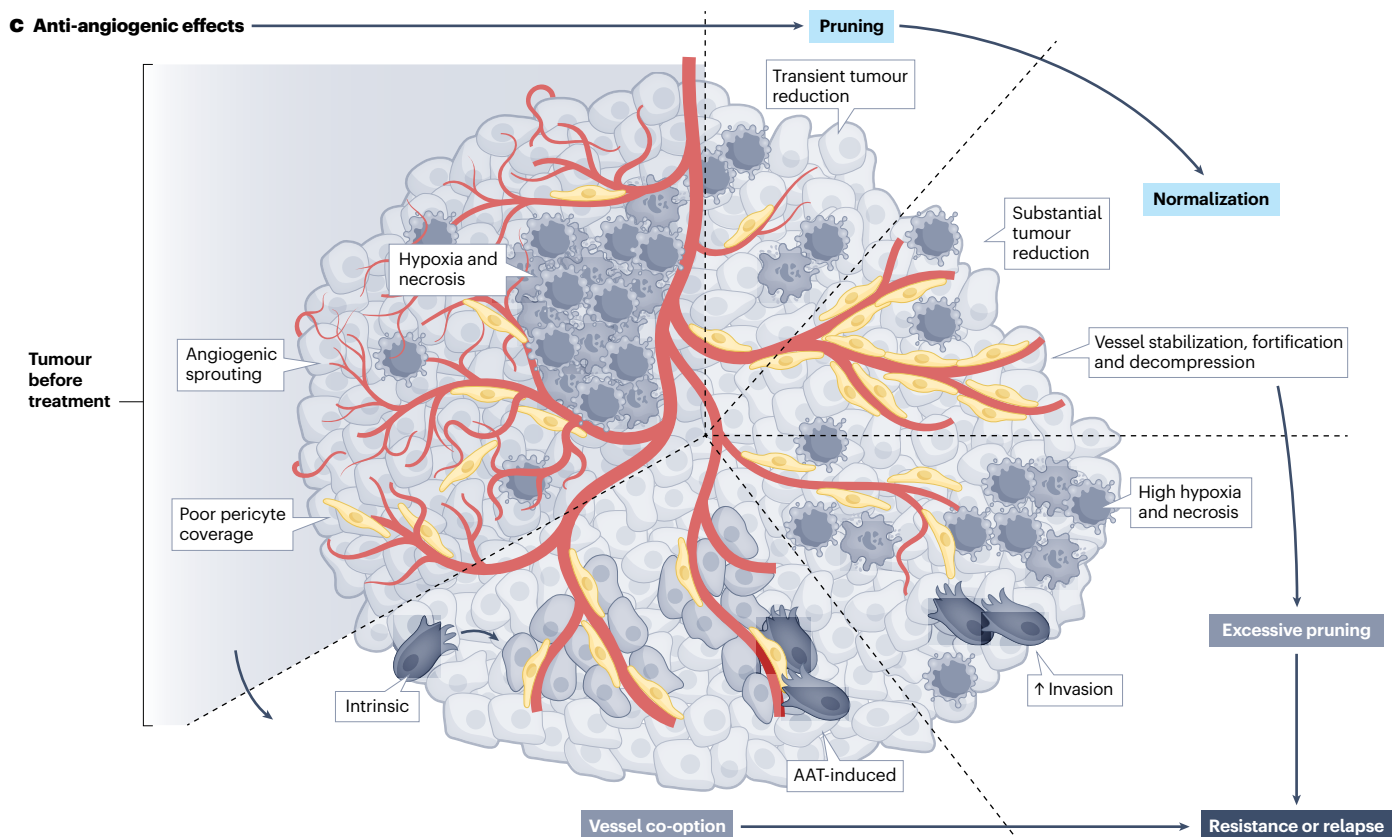
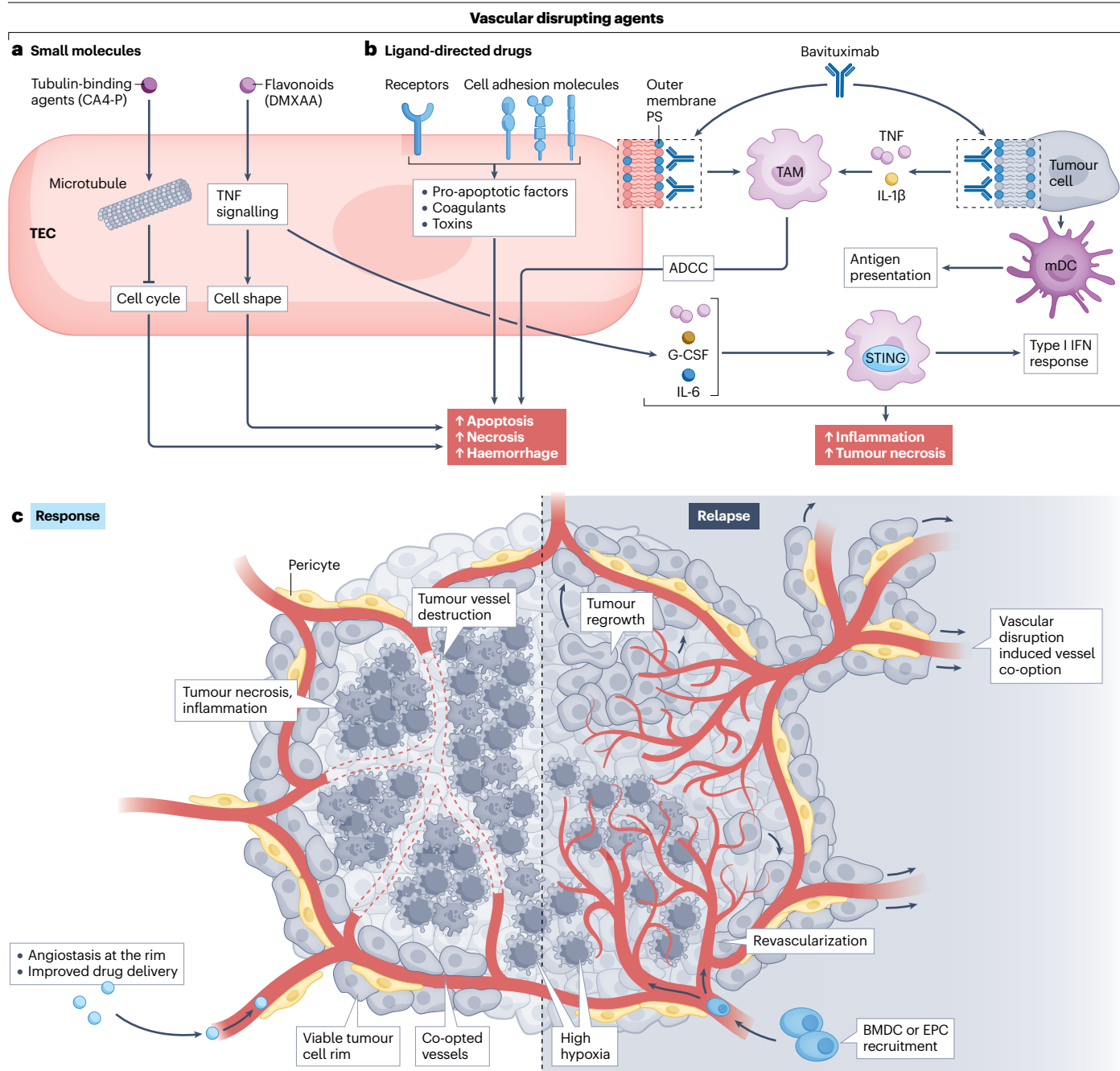


Fig. 1 | Anti-angiogenic strategy: the beginning of vascular targeting therapies. Anti-angiogenic therapies (AATs) were developed with the intention of trimming the tumour vascular tree, which is abnormally composed of angiogenic sprouts formed from pre-existing vessels. **a**, Anti-angiogenic targets include $\alpha V\beta 3$ or $\alpha V\beta 5$ integrins, angiopoietins, Notch ligands and mostly tyrosine kinase signalling, such as the vascular endothelial growth factor (VEGF)–VEGF receptor (VEGFR) axis. These targets control major signalling pathways involved in endothelial cell proliferation, migration and survival, ultimately triggering vessel sprouting. **b**, The tumour growth stage and state of vascularization are key factors for effective targeting during the AAT window. **c**, Upon AAT, sprouting inhibition induces vessel pruning leading to transient tumour starvation and

growth reduction. Fine-tuning of AAT leads to normalization associated with stabilized, fortified and decompressed tumour vessels. However, excessive pruning increases necrotic and hypoxic tumour areas, triggering AAT-induced resistance mechanisms whereby evading tumour cells co-opt, proliferate and invade along normal blood vessels, ultimately leading to metastasis. Vessel co-option has also been demonstrated as an intrinsic alternative mode of neovascularization in several tumour types, rendering AAT ineffective. ANG2, angiopoietin-2; DLL4, delta-like ligand 4; FGF, fibroblast growth factor; FGFR, FGF receptor; JAG1, JAGGED 1; PDGF, platelet-derived growth factor; PDGFR, PDGF receptor; TGF β , transforming growth factor β ; TGF β R, TGF β receptor; TIE2, tyrosine kinase with immunoglobulin and EGF homology domains 2.



cytotoxic drug vinblastine, which had stabilizing effects in mice bearing neuroblastoma for a transient period, as did the VEGF receptor 2 (VEGFR2)-blocking antibody DC101, but combined and continuous treatment was able to regress tumours for a long-lasting time period. More recent reports describe multiple examples of how inflammation is induced with MTDs but that metronomic chemotherapy suppresses inflammation, thus providing exciting potential benefits for combination with immunotherapies^{56–59}.

Despite exciting and rapid advances in many preclinical models around the time of the initial approvals, the clinical translation of anti-angiogenic agents was still relatively limited⁶⁰. Part of the reason for this included that the majority of preclinical mouse models were

restricted to small and fast-growing cancers that lacked salient features of human cancer such as the complexity of blood vessel heterogeneity and maturity. This problem became most apparent when testing anti-angiogenic agents in spontaneous, orthotopic mouse models of cancer, in which enduring cancer control was not achieved, and sometimes treatment even evoked enhanced invasion or metastasis^{61–64}. In line with these observations, clinical studies revealed that the efficacy of VEGF or VEGFR2 inhibitors was restricted to only a subset of cancer types with transitory beneficial effects leading to progression-free survival (PFS) but not overall survival (OS) benefits^{65–70}. Several reasons were proposed for why vessel abrogation with anti-angiogenic agents was insufficient to halt tumour growth⁷¹. Despite the substantial vessel

Fig. 2 | Vessel destruction. Vascular disruption specifically targets tumour endothelial cells (TECs) based on their distinct phenotype compared with their physiological counterparts. **a**, Small molecular vascular disrupting agents (VDAs) comprise tubulin-binding agents (combretastatin A4 phosphate (CA4-P)) and flavonoids (5,6-dimethylxanthenone-4-acetic acid (DMXAA)) that alter microtubule stability and cytoskeleton organization, thus affecting cell cycle dynamics and cell morphogenesis and leading to the specific apoptosis of TECs. DMXAA also triggers the secretion of tumour necrosis factor (TNF)-dependent pro-inflammatory cytokines such as granulocyte-colony-stimulating factor (G-CSF) and IL-6 into the tumour microenvironment. This induces a cooperative pro-inflammatory type I interferon (IFN) response via the stimulator of interferon genes (STING) pathway involving anti-tumoural tumour-associated macrophages (TAM_{Anti}), monocytes and T cells. **b**, Ligand-directed VDAs target specific receptors and cell adhesion molecules of TECs to deliver coagulants,

toxins or pro-apoptotic factors that induce TEC cytotoxicity. Bavituximab is a chimeric monoclonal antibody with immunomodulatory properties, which binds phosphatidylserine (PS) exposed on the outer membrane of TECs but absent on normal blood vessels¹²⁷. This activates an antibody-dependent cellular cytotoxicity (ADCC) pro-inflammatory response of TAM_{Anti}, which leads to TEC apoptosis^{128,129}. Bavituximab also binds the outer membrane PS on tumour cells, further amplifying the pro-inflammatory response of TAM_{Anti} via TNF and IL-1β¹³⁰ and additionally inducing dendritic cell (DC) maturation and antigen presentation¹²⁸. **c**, VDAs cause massive apoptosis, necrosis and haemorrhage of the tumour vasculature, further triggering an antitumour immune response and tumour necrosis. However, the low oxygen levels in the necrotic core will inevitably trigger tumour re-growth and relapse in part due to the recruitment of pro-angiogenic bone-marrow-derived cells (BMDCs) and endothelial progenitor cells (EPCs) and escaped tumour cells at the rim that co-opt blood vessels. mDC, mature dendritic cell.

abnormalities observed in several angiogenic tumours, only 4–8% of TECs proliferate as recently confirmed by single-cell transcriptional profiling of human and mouse tumours^{72,73}. As most anti-angiogenic agents inhibit the early stages of angiogenesis including EC proliferation, migration and sprouting, it became clear that early-stage tumours were more responsive to anti-angiogenic therapies than established late-stage tumours^{17,74,75} (Fig. 1b). Furthermore, during anti-VEGF therapy, remaining tumour vessels underwent fortification with an improved pericyte coverage counteracting the therapeutic effects⁷⁶ and highlighting the impact of tumour vessel heterogeneity on therapeutic responses^{73,77–79} (Fig. 1c). VEGF inhibitors also increased the expression of gene signatures associated with mature homeostatic vascular functions indicative of a maturing tumour vasculature and resistance to anti-angiogenic approaches⁷². Subsequently, strategies were tested to remove pericytes^{80,81} from established tumour vessels, predominantly by inhibiting PDGF receptor (PDGFR) and thus inducing sensitization to anti-VEGF agents^{76,82}. This strategy had prolonged beneficial effects in preclinical models but not necessarily in the clinical setting, although sunitinib and sorafenib, both capable of blocking VEGFR and PDGFR, but not the VEGF antibody bevacizumab, were FDA-approved for renal cell carcinomas and hepatocellular carcinomas, respectively^{83,84}.

It is important to note that VEGF, FGF2 and other pro-angiogenic growth factors rely on downstream metabolic adaptations to induce vessel sprouting, which emerged from the observation that TECs undergoing an angiogenic switch also become metabolically rewired by increasing glycolysis, fatty acid and glutamine metabolic pathways. Although tip cells appear to rely more on a 6-phosphofructo-2-kinase/fructose-2,6-bisphosphatase (PFKFB3)-driven glycolytic metabolism to produce enough ATP for vessel sprouting, stalk cells depend on fatty acid oxidation for vessel sprout elongation. This has led to preclinical targeting of PFKFB3 or carnitine palmitoyltransferase 1A (involved in fatty acid metabolism) eliciting reduced angiogenesis and subsequent tumour growth^{85,86}. Nevertheless, as this metabolic strategy also blocks angiogenic sprouting, resistance mechanisms similar to those of the upstream pro-angiogenic receptor tyrosine kinase signalling cues are likely to occur.

Finally, another reason why anti-angiogenic therapies have been less successful in the clinical setting likely lies in non-angiogenic, alternative modes of vascularization. These include vasculogenic mimicry, whereby tumour cells, in the absence of an EC lining, differentiate and align to acquire blood vessel functions^{87,88}, and vascular co-option, where tumour cells align to established blood vessels⁸⁹

(Fig. 1c and Box 1). As these processes are not stimulated by the targets of anti-angiogenic therapy, they would be unresponsive to anti-angiogenic treatments.

Vascular disrupting strategies

Given the limitations of anti-angiogenic therapy, alternative vascular disrupting agents (VDAs) were developed to specifically disrupt highly unstable and, thus, more sensitive tumour vessels^{90,91}. Small molecular VDAs comprising tubulin-binding agents and flavonoids induce TEC apoptosis and necrosis^{92,93} (Fig. 2a). Tubulin-binding agents such as combretastatins depolymerize and disorganize the TEC cytoskeleton, altering their shape, motility, adhesion, proliferation and function^{94,95} (Fig. 2a). Flavonoids are plant derivatives with anticancer properties, which also exert vascular cytotoxicity through a pro-inflammatory cascade^{96,97}. Ligand-directed molecules are the second class of VDAs that deliver toxins, coagulants or pro-apoptotic factors selectively to TECs, via binding to specific surface proteins including integrins, cell adhesion molecules and receptors^{98–100} (Fig. 2b). Irrespective of the VDA type, damaged TECs induce haemorrhagic necrosis, reduce blood flow and block oxygen and nutrient transport, ultimately shutting down tumour growth. Given their efficacy in preclinical studies, the small molecular VDAs combretastatin A4-phosphate and the flavonoid 5,6-dimethylxanthenone-4-acetic acid (DMXAA) were evaluated clinically for a wide range of tumour types reaching phase III trials but failed to show significant improvement of PFS or OS in patients^{92,101–103}. The limitations of VDAs identified in clinical trials to date include cardiovascular toxicity¹⁰⁴ and a lack of effect at the tumour rim likely owing to co-opted normal blood vessels that are unaffected by VDAs¹⁰⁵. In addition, high levels of hypoxia at the necrotic tumour core and the recruitment of endothelial progenitor cells and bone-marrow-derived monocytes induce neovascularization and immunosuppression, unequivocally causing resistance and recurrence^{106,107} (Fig. 2c).

To date, clinical trials testing VDAs together with chemotherapy have had mixed results, supporting the additive effect of VDAs and hinting that careful treatment scheduling could potentiate chemotherapeutic effects and target the tumour cells at the rim^{104,108–110}. Although combinations of combretastatins with anti-angiogenics to more efficiently block re-neovascularization showed tumour reduction and improved survival in mouse models^{94,111,112}, no significant beneficial effects were observed in phase II clinical trials^{113,114}. Other recent studies combined combretastatins with immunotherapies with positive outcomes in mammary and hepatocellular carcinoma mouse models^{115,116}. In line with this, combinations of DMXAA with the hypoxia inhibitor

digoxin¹¹⁷ or with chemotherapy and/or immunotherapy within co-delivery systems^{118–123} boosted antitumour immunity^{124–126}, in addition to antitumour and antivascular effects¹⁰¹. Similarly, bavituximab, which was originally described as a VDA targeting phosphatidylserines exposed on TECs^{127,128}, also exerts antitumour activity and reverses immunosuppression^{129,130} (Fig. 2b) when combined with chemoradiotherapy or with programmed cell death protein 1 (PD1) immune checkpoint blockade (ICB)^{131–133}. Thus, using VDAs in combination therapy requires additional preclinical investigation and clinical trials to assess its relevance, especially with immunomodulatory therapies in which encouraging results have been recently observed.

Vascular promotion

One of the most provocative and counterintuitive vascular-targeting concepts to alleviate hypoxia and improve drug delivery is the notion of ‘vascular promotion therapy’, by which blood vessel growth is fostered or promoted. In preclinical mouse models of cancer, the concept was tested by exploiting the pro-angiogenic properties of low-dose cilengitide, a cyclic RGD mimetic targeting $\alpha V\beta 3$ and $\alpha V\beta 5$ integrins, which at low doses enhances VEGF-stimulated angiogenesis by upregulating VEGFR2 recycling in ECs^{134–136}. Combining low-dose cilengitide with the calcium channel blocker that has vasodilation properties, verapamil, and the chemotherapeutic agent gemcitabine provided a dual action by enhancing blood vessel density, blood flow and gemcitabine delivery, while reducing pericyte coverage and tumour hypoxia, in addition to increasing the metabolism of gemcitabine by altering gemcitabine transporter expression and killing activity in malignant cells. This approach prevented the growth of pancreatic ductal adenocarcinoma and NSCLC tumours and metastasis in mice while imparting reduced toxic effects from gemcitabine^{136–138}.

More recently, combinations of low-dose cilengitide with the chemotherapeutic agent doxorubicin have been delivered using endothelial-specific liposomes, demonstrating similar beneficial effects *ex vivo* and *in vivo* with an improved delivery specificity¹³⁹. Although cilengitide is off-patent because of its failure in clinical trials at MTDs, these data raise the opportunity for further investigation to repurpose it for use at low doses in combination with standard of care. Furthermore, new orally available RGD-mimetic prodrugs have been developed¹⁴⁰ and testing these in combination with standard-of-care therapies may provide additional alternatives (Fig. 3).

Since this initial proposition, new combinations of vascular remodelling agents in combination with other chemotherapies have proven successful as a vascular promotion therapy. For instance, eribulin mesylate, an inhibitor of microtubule dynamics, is a clinically relevant candidate with strong antitumour and provascular activities^{141,142}. Combinations with either capecitabine or paclitaxel increased blood vessel density, reduced hypoxia, improved antitumour immunity and, more importantly, controlled tumour growth in breast cancer and clear cell sarcoma mouse models^{143,144}. Lysophosphatidic acid (LPA), which is a central bioactive phospholipid in cancer¹⁴⁵, can also promote the formation of fine and permeable vascular networks in subcutaneous preclinical models of lung, melanoma and colon cancers¹⁴⁶. When used in combination with 5-fluorouracil or oxaliplatin, tumour growth was suppressed¹⁴⁶. Mechanistically, LPA improved drug delivery via the specific activation of the LPA4 receptor, which promotes the formation of the linear adherens junction by vascular endothelial-cadherin in TECs, ultimately enhancing vascular network formation¹⁴⁶. Similar to verapamil¹³⁷, nitric oxide is another interesting ‘vascular promoting’ candidate with vasodilating effects among many other pleiotropic

vascular functions^{147,148}. When co-delivered with paclitaxel using a specific polymeric hybrid micelle system, pro-angiogenic effects, evidenced by increased vessel density, perfusion and permeability, and synergistic antitumour effects, evidenced by efficient drug accumulation and reduced tumour growth, were observed in several mouse tumour models¹⁴⁹ (Fig. 3). Thus, although not yet tested in humans, these reports suggest an alternative to anti-angiogenic or VDA therapy, whereby increasing blood vessel numbers and function could provide benefit as a cancer treatment.

Targeting paracrine signalling

The undesired tumour adaptations induced by tumour vessel destruction raised the question of whether targeting EC signalling cues, while maintaining the vessels themselves, could be an effective strategy for cancer therapy. These paracrine cues not only generate chemical gradients and niches that affect immune cell infiltration and tumour cell responses to therapy locally but also have systemic effects, which may have important implications in drug resistance¹⁵⁰.

Angiocrine factors support organogenesis^{151–153}, maintain homeostasis in organ-specific vascular beds and promote regeneration and repair^{27,154–156}. As cancer shares features of embryogenesis and regeneration, an ‘angiocrine switch’ has been proposed whereby activated TECs establish a pro-tumoural vascular niche rich in growth factors, trophogens, cytokines, chemokines and adhesion molecules that sustain tumour growth¹⁵⁷. The roles of angiocrine factors in cancer fall broadly into four functionally overlapping regulatory groups: (1) stem cell and progenitor cell proliferation; (2) tumour dormancy and pre-metastatic niche formation; (3) tumour growth, invasiveness and chemoresistance; and (4) tumour–immune cell crosstalk and immune tolerance¹⁵⁸ (Fig. 4). Among these angiocrine factors, EC-derived JAGGED1 (a Notch ligand) and subsequent downstream Notch activity have been shown to be involved in several angiocrine mechanisms in tumours, including the control of tumour cell survival in colorectal and breast cancer^{159,160} and the promotion of malignant cell proliferation and EMT in mouse models of lung and prostate cancer¹⁶¹. Effects of altered angiocrine signalling in metastasis regulation and cachexia have also become apparent^{12,162}. The communication triad of the immune cell, malignant cell and EC was shown to be crucial at pre-metastatic sites whereby it led to the activation of ECs and the promotion of a pro-inflammatory microenvironment via vascular cell adhesion molecule 1 (VCAM1), ultimately facilitating the adhesion of malignant cells to the vessel wall, their extravasation and metastasis¹⁶³ (Fig. 4).

Angiocrine factors also have a considerable role in chemoresistance and handling genotoxic stress. For example, doxorubicin, which is used in the first-line treatment of Burkitt’s lymphoma, induces secretion of IL-6 from ECs, generating minimal residual disease that serves as a reservoir for cancer relapse¹⁶⁴. In a subsequent study, including mouse models of lung and skin cancer, the EC-specific genetic deletion of focal adhesion kinase (FAK) revealed a mechanism by which this regulation of EC-derived IL-6 was controlled following DNA-damaging therapies including doxorubicin or radiation therapy. Specifically, deletion of FAK in ECs in established tumours *in vivo* was sufficient to enhance the cytotoxic effects of DNA-damaging agents by altering the cytokine angiocrine signal and making tumour cells more vulnerable to DNA damage¹⁶⁵. Similarly, loss of FAK in ECs was shown to be responsible for reduced metastasis and increased survival of mice bearing pancreatic ductal adenocarcinoma tumours and treated with gemcitabine owing to altered angiocrine signalling¹⁶⁶. Such phenotypes likely involve the kinase activity of

Vascular promotion strategy

Vascular remodelling agents and chemotherapy

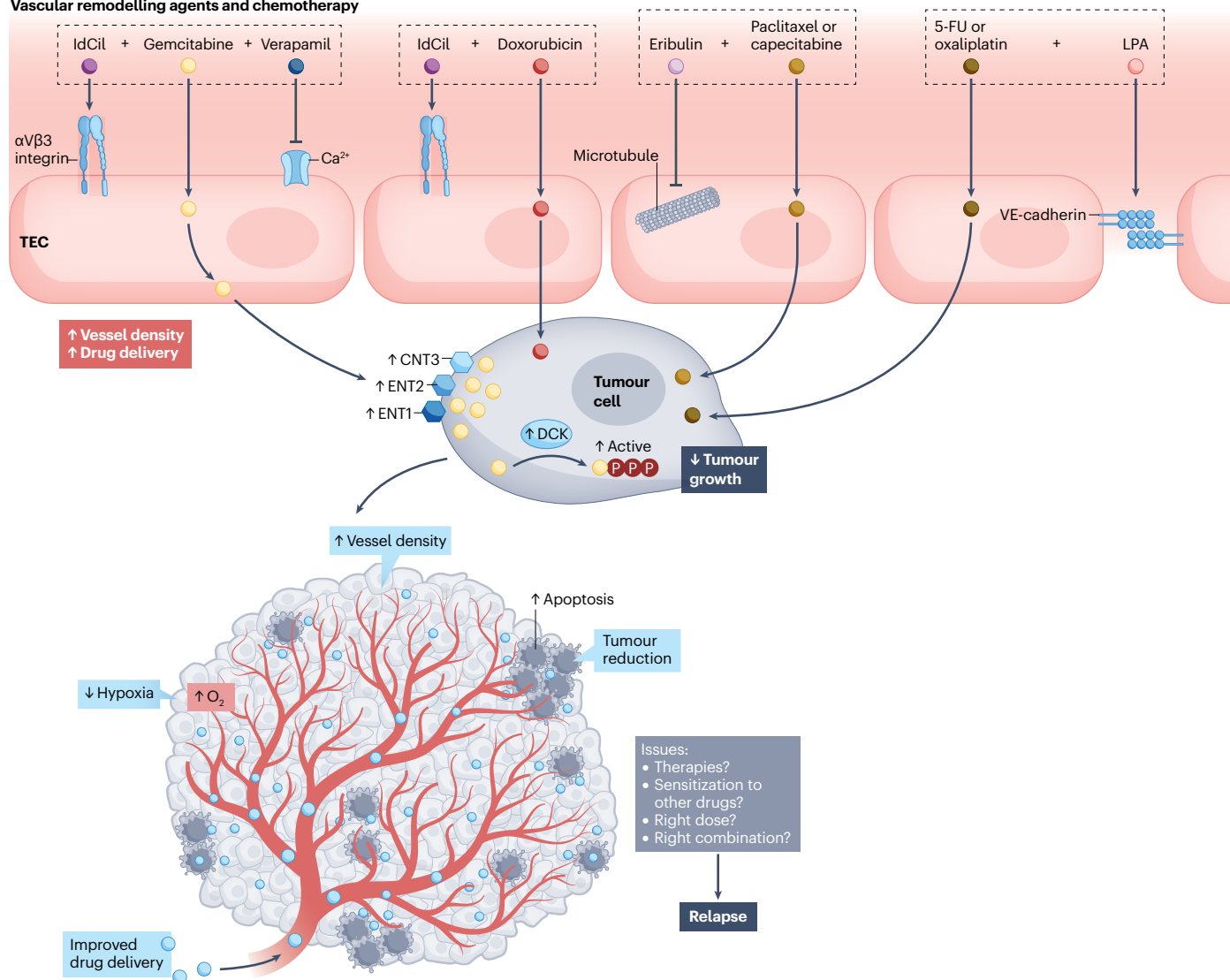


Fig. 3 | Vascular promotion therapy. The basis of this strategy is the promotion of new, more functional blood vessels to improve blood supply to tumours. Combining specific doses of vascular remodelling agents with chemotherapy was shown to increase vessel density, perfusion and oxygenation, thereby potentiating the efficacy of chemotherapy treatment. Specifically, low doses of cilengitide (IdCil), an RGD mimetic, which blocks $\alpha V\beta 3$ integrin adhesive function at high doses but not at low doses, combined with the calcium channel blocker and vasodilator verapamil, enhance vascular density and thus gemcitabine delivery to significantly reduce tumour growth. This combination sensitizes tumour cells to chemotherapy by promoting the uptake of gemcitabine via transporter proteins equilibrative nucleoside transporter 1 (ENT1), ENT2 and concentrative nucleoside transporter 3 (CNT3) and by promoting the conversion of gemcitabine to its active intracellular form via the

upregulation of deoxycytidine kinase (DCK). Similar results were observed using membrane type 1-matrix metalloproteinase (MT1-MMP)-activated cilengitide modified onto liposomes loaded with the chemotherapy doxorubicin (known as MC-T-Dox). Another combination of eribulin, an inhibitor of microtubule dynamics with paclitaxel or capecitabine, has shown similar effects both in vitro and in vivo. Finally, lysophosphatidic acid (LPA) used in combination with either 5-fluorouracil (5-FU) or oxaliplatin increases vessel density and improves drug delivery to tumours, via membrane localization of vascular endothelial-cadherin (VE-cadherin) in tumour endothelial cells (TECs). Further studies will determine whether this strategy is applicable in cancer therapy with the right dose or combination, whether other cytotoxic drugs can be sensitized in a similar manner and finally whether the strategy causes relapse. O₂, oxygen.

EC-FAK¹⁶⁷ and are supported by evidence showing an association between reduced p397FAK expression (an indicator of activated FAK) in ECs and improved survival in patients with neoadjuvant-treated locally advanced breast cancer¹⁶⁸.

Chemoresistance can also be initiated by the malignant cell compartment, which, after crosstalk with surrounding ECs via angiocrine signals, further promotes this resistance in a feedforward loop. High levels of CXC-chemokine ligand 1 (CXCL1) and CXCL2 in carcinoma

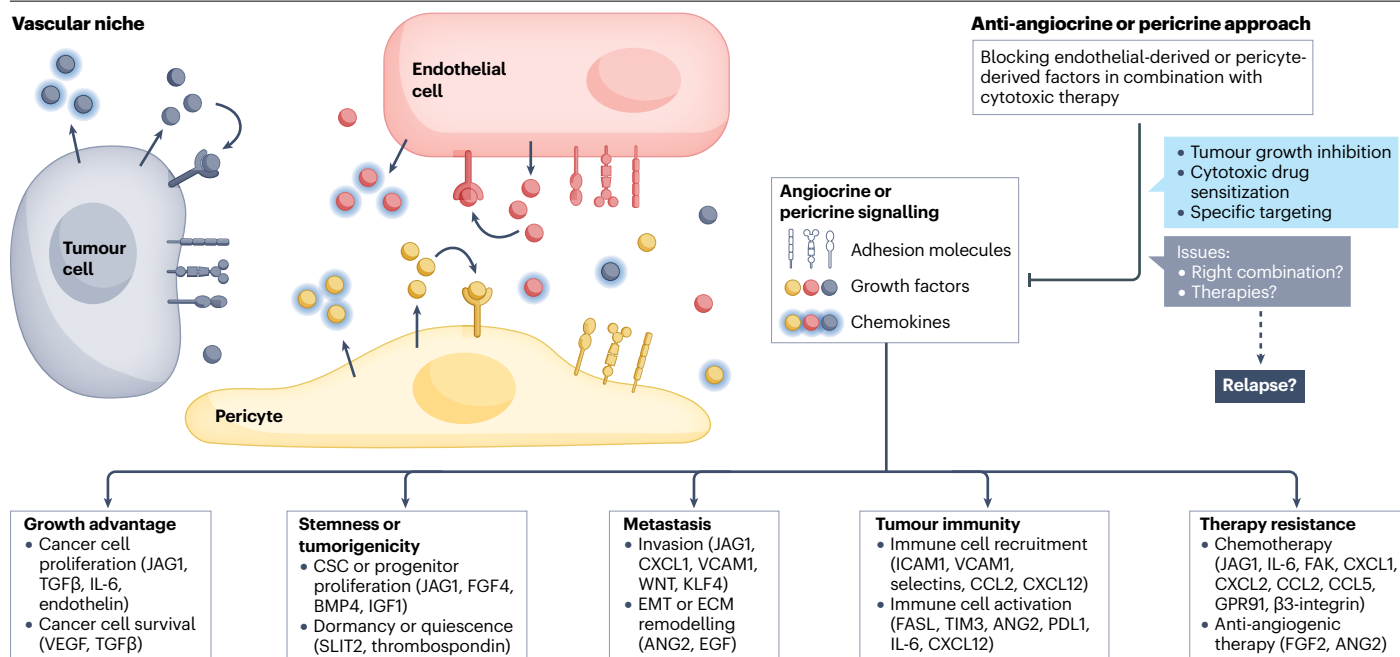


Fig. 4 | Targeting endothelial-derived and pericyte-derived factors.

This approach is designed to specifically target endothelial-derived and pericyte-derived factors that promote tumour advantages within the vascular tumour niche. When used in combination, this would ‘reprogram’ and sensitize tumours to cytotoxic drugs. Potential targets include active growth factors, which are secreted in an autocrine and paracrine fashion, adhesion molecules regulating tumour and immune cell trafficking through blood vessel walls, and chemokines controlling the migration of tumour and immune cells. These factors confer growth advantages to tumour cells by promoting their proliferation (JAGGED1, transforming growth factor β (TGF β), IL-6 and endothelin) and survival (vascular endothelial growth factor (VEGF) and TGF β). They also control stemness and tumorigenicity by increasing the proliferation of cancer stem cells (CSCs) or progenitors (JAGGED1 (JAG1), fibroblast growth factor 4 (FGF4), bone morphogenetic protein (BMP4) and insulin-like growth factor 1 (IGF1)) and by inducing dormancy and quiescence (SLIT2 and thrombospondin).

In addition, they participate in tumour cell metastasis via the regulation of invasion (JAG1, CXCL1, VCAM1, WNT and Krüppel-like factor 4 (KLF4)) and epithelial-to-mesenchymal transition (EMT) or extracellular matrix (ECM) remodelling (angiopoietin-2 (ANG2) and epidermal growth factor (EGF)). Furthermore, they potentiate antitumour immunity by controlling immune cell recruitment (intercellular adhesion molecule 1 (ICAM1), VCAM1, selectins, CC-chemokine ligand 2 (CCL2) and CXCL12) and activation (FAS ligand (FASL), T cell immunoglobulin domain and mucin domain 3 (TIM3), ANG2, programmed cell death protein 1 ligand 1 (PDL1), IL-6 and CXCL12). Finally, they are key players in the resistance to conventional anticancer therapies, including chemotherapy (JAGGED1, IL-6, focal adhesion kinase (FAK), CXCL1, CXCL2, CCL2, CCL5, G protein-coupled receptor 91 (GPR91; also known as succinate receptor 1) and β 3-integrin) and anti-angiogenic therapies (ANG2 and FGF2).

cells induce an increase in tumour necrosis factor (TNF) production in ECs, further enhancing CXCL1 and CXCL2 to recruit CD11B⁺GR1⁺ myeloid cells, ultimately promoting malignant cell survival and thereby resistance to therapy¹⁶⁹. Similarly, lymphoma cells increase their aggressiveness via FGF4–FGF receptor 1 signalling upstream of JAGGED1–NOTCH2 activation in neighbouring ECs¹⁷⁰. In another example, in mouse cancer models and patient-derived xenograft models, cytotoxic stress enhances expression of the ETS2 transcription factor in ECs via FGF4–FGF receptor 1 activation, inducing insulin-like growth factor 1 expression and reducing insulin-like growth factor-binding protein 7 levels in ECs, which lead to expansion of the chemoresistant cancer stem cell niche¹⁷¹.

Studies have also shown that paracrine signalling cues emanating from pericytes, coined pericyte cues¹⁰, can affect tumour growth and metastatic dissemination^{172–175}. For example, genetic ablation of pericyte FAK or β 3-integrin enhanced tumour growth via regulation of growth arrest-specific protein–AXL and CC-chemokine ligand 2 (CCL2) pathways in mice^{10,176,177}. Furthermore, low levels of mural cell β 3-integrin in tumours were found to be associated with

increased tumour burden and progression in patients with oestrogen receptor positive breast cancer, cutaneous melanoma and large B cell lymphoma without affecting blood vessel numbers or conduit functions¹⁰.

Angiocrine signals, however, can also exert suppressive functions on tumour cells. EC-derived SLIT2 is an angiocrine factor that is repressed by high levels of ephrin receptor 2A in breast cancer, which in turn facilitates breast tumour growth¹⁷⁸. EC-derived thrombospondin also induces tumour cell quiescence in mouse and zebrafish models of breast cancer¹⁷⁹. Moreover, perlecan-expressing ECs inhibit breast and lung carcinoma growth by inhibiting both the pro-tumorigenic and pro-inflammatory signals derived from cancer cells. Inhibition of perlecan expression in ECs rescued this tumour suppressive response by inducing an increase in IL-6 production by the cancer cells¹⁸⁰. Notably, newer reports also support a role for lung EC-derived WNT signalling in regulating metastasis, with spatial vascular molecular mapping revealing angiocrine TIE1 receptor signalling as a specific regulator of vascular WNT activity^{12,181} (Fig. 4).

Vessel normalization

Vascular normalization has been proposed¹⁸² as a process whereby initial effects of anti-angiogenic therapy cause temporary pruning, fortification and decompression of tumour blood vessels^{183,184} and provide a transient therapeutic window for improving the efficacy of radiation and cytotoxic therapy while reducing hypoxia and high interstitial pressure^{185,186}. It is important to note, however, that the tumour vasculature does not become normal, but the aberrant effects emanating from the dysregulated signals are partly reverted. As factors that control vessel normalization have been covered previously in multiple outstanding reviews^{186–189}, only their targets and overall effects are illustrated in Fig. 5. Most importantly, vessel normalization is a transient process until continuous vessel pruning and reduction culminate in intratumoural hypoxia, leading to aggressive tumour relapse (Fig. 1). This was demonstrated in an early patient study in which bevacizumab treatment induced a rapid decrease in chemotherapy delivery indicating the importance of dosing and scheduling

to achieve robust benefits¹⁹⁰. Moreover, high-dose anti-VEGFR2 or VEGFR2 inhibitor treatment in multiple breast cancer mouse models revealed an adaptive resistance mechanism in which IL-17A expression in $\gamma\delta$ T cells and N2-like neutrophil polarization was promoted, which in turn facilitated CD8⁺ T cell exhaustion and an immunosuppressive microenvironment^{191,192}. However, this effect was not observed with low-dose anti-VEGFR2 regimens. These results are in line with several investigations, in which substantially lower doses of VEGF inhibitors prolonged vascular normalization in several mouse tumour models without excessive vessel pruning^{193,194}. This led to the idea of lowering the dose and reducing the number of rounds of treatment with angiogenesis inhibitors to maintain a normalization window for a prolonged time period and avoid the effects of high concentrations of VEGF inhibitors⁷⁵.

Similarly, impeding TEC expansion with a low dose of inhibitors that target glycolysis and fatty acid oxidation may be another promising option as ECs exhibit a greater sensitivity to these inhibitors

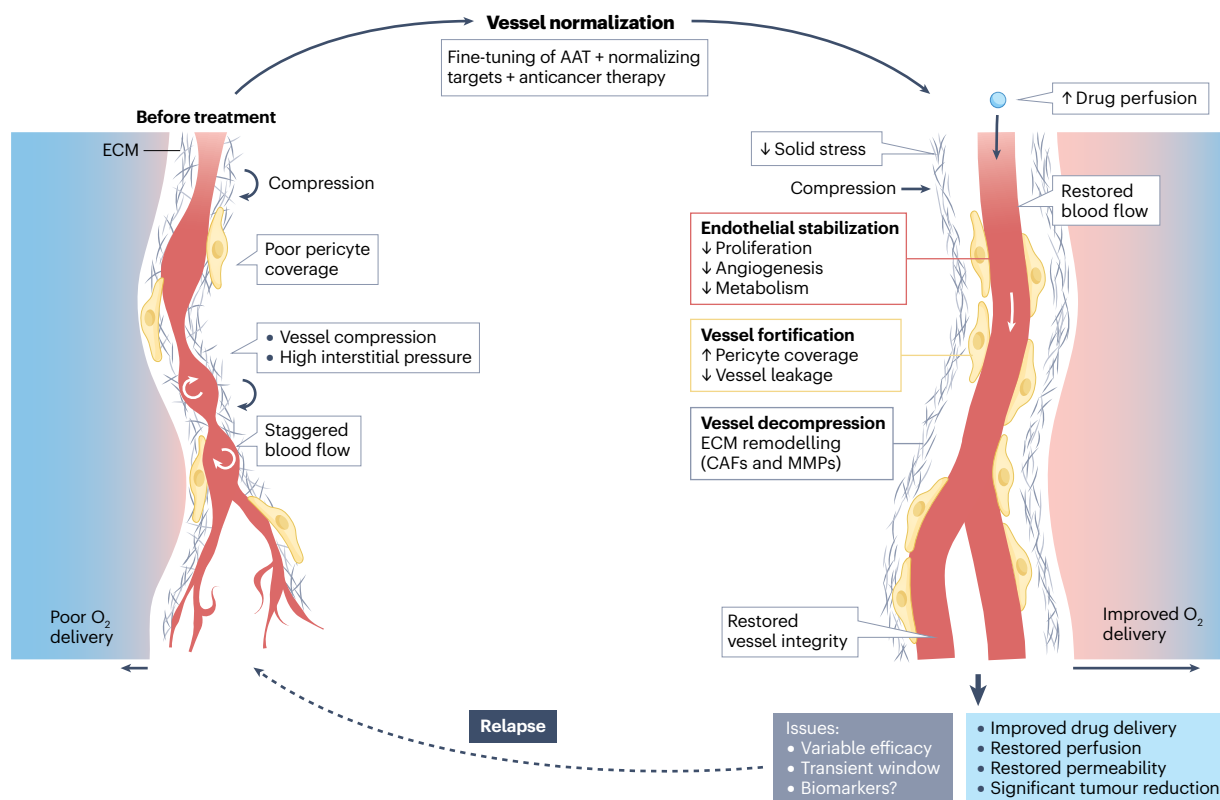


Fig. 5 | Vessel normalization. The concept of vessel normalization proposes to alleviate the ineffectiveness of single anti-angiogenic therapy (AAT) treatment regimens by fine-tuning of AAT administration in combination with specific ‘normalizing’ agents and cytotoxic cancer therapies. The overarching goal of this approach is to target the right window of opportunity during tumour and vessel growth to restore proper vessel functionality. Normalization of the tumour vasculature requires not only reduced sprouting of tumour endothelial cells upon anti-vascular endothelial growth factor (VEGF) or anti-vascular endothelial growth factor receptor (VEGFR) therapy but also stabilization of tumour endothelial cells towards a quiescent phenotype. For instance, this can be achieved by inhibition of pro-angiogenic and pro-proliferative signals or specific targeting of endothelial metabolism. Furthermore, vascular integrity should be restored owing to enhanced fortification by pericytes and smooth muscle cells,

for example, via re-establishment of a homogeneous basement membrane or inhibition of pro-proliferative pericyte-specific signals. Importantly, vessel decompression is key to alleviating solid stress and interstitial fluid pressure of leaky angiogenic tumour vessels and to restoring proper blood flow, vessel permeability and oxygenation. For instance, this can be achieved by targeting extracellular matrix (ECM) remodelling by cancer-associated fibroblasts (CAFs) or by directly targeting matrix metalloproteinases (MMPs). Overall, vessel normalization restores perfusion and improves cytotoxic drug delivery, inducing substantial tumour reduction and providing beneficial effects for patients. However, this efficacy is variable depending on the tumour type, with only a transient window for effective targeting and a lack of biomarkers to validate this efficacy, eventually leading to tumour relapse. O₂, oxygen.

than cancer cells^{195,196}. Although higher doses of a PFKFB3 blocker caused tumour vessel disintegration by substantially reducing glycolysis, a low dose that decreased endothelial glycolysis by only 15–20% normalized the tumour vasculature, reduced metastasis and improved chemotherapy without affecting the cancer cells⁸⁵. TECs from patients with colorectal cancer exhibit enhanced glycolysis with elevated glyceraldehyde-3-phosphate dehydrogenase expression compared with ECs from adjacent peritumoural tissues¹⁹⁷. Low-dose treatment with osimertinib, which is an inhibitor of the epidermal growth factor receptor, caused tumour regression with vascular normalization and increased the infiltration of immune effector cells in mice bearing colorectal MC38 and CT2A tumours; this was due to the reduced secretion of lactate from TECs as osimertinib was able to inhibit glyceraldehyde-3-phosphate dehydrogenase¹⁹⁷.

To date, lowering the dose of bevacizumab appears to have superior effects in patients with glioblastoma compared with the conventional higher dose^{198–200}. Encouraging effects of weekly, low-dose administration of bevacizumab were also recently reported for patients with platinum-resistant ovarian carcinoma²⁰¹. In addition, favouring the concept of ‘less is more’ in phase II clinical trials, bevacizumab in combination with low-dose metronomic oral cyclophosphamide was also shown to be efficacious in patients with recurrent ovarian cancer^{202,203}. Contrary to these findings, however, are some reports indicating that vascular normalization in mouse models of glioblastoma is associated with an increase in hypoxia or no change, thus clearly an improved understanding of the consequences and control of vascular normalization is still required²⁰⁴.

A major hurdle to the robust success of vascular normalization is the lack of reliable predictive biomarkers of response to accurately determine a therapeutic window. Currently, these are still limited to measuring blood perfusion and oxygenation^{205,206}, as evidenced in glioblastoma^{207–209}, NSCLC²¹⁰ and breast cancer^{211,212}. Finally, the adaptive mechanisms orchestrated by the tumour to counteract and resist the therapy are difficult to overcome^{63,192}. Apart from these challenges, not all cancer types and stages are susceptible to vascular normalization²⁰⁶, especially those with high angiopoietin-2 (ANG2) levels that destabilize tumour vessels²¹³ and in the early stages of metastasis in which vessel co-option is prominent (Box 1).

Angiogenesis and immunosuppression

The effects of all treatment strategies described earlier seem to be dependent on one major common denominator, that is, the degree of vessel functionality. A wealth of research has provided unambiguous agreement that excessively depleting or even abrogating tumour vessels leads to a fast relapse creating a potentially even more aggressive tumour, which results from therapy-induced hypoxia-instigated mechanisms to reinvigorate tumour growth^{214,215}. The consequences are multifold; hypoxic and necrotic tumours enhance cell stemness, become genetically unstable and either try to reinstate vessel growth to obtain sufficient oxygenation or find alternative ways to grow and expand by metabolic adaptation or by EMT induction stimulating cell migration^{216,217} (Fig. 1). These tumours also become more resistant to standard and immune therapies.

Delving further into the underlying mechanisms, one of the most prominent traits that limit anti-angiogenic therapy in cancer is hypoxia-induced immunosuppression^{218–220}. Indeed, what these studies also revealed is a reciprocal and intricate relationship between vascular and immune cells by which both cell compartments work together to generate an overall angiogenic and immunosuppressive

phenotype in cancer^{221,222} (Fig. 6), similar to their behaviour during wound closure^{23,223}. This phenotype is exacerbated when tissue oxygen levels become scarce, leading to the infiltration of chemokine-secreting immune cells, including macrophages, tyrosine kinase with immunoglobulin and EGF homology domains 2 (TIE2; also known as angiopoietin-1 receptor)⁺ monocytes, myeloid-derived suppressor cells (MDSCs), neutrophils and regulatory T (T_{reg}) cells that contribute to immune tolerance and re-neovascularization in tumours partly via the secretion of cocktails of pro-angiogenic and immunosuppressive factors^{224–226}. This includes VEGF, FGF1 and FGF2, prokineticin-2 (PK2, also known as Bv8), TNF, transforming growth factor β , IL-10, PDGFB, placental growth factor (PIGF), transmembrane proteins such as neuropilin-1 and various CXCLs, as described in various studies (reviewed elsewhere^{218,227,228}). Interestingly, the elevated levels of some of these angiogenic molecules could also be detected in the peripheral blood of patients undergoing anti-angiogenic therapy. For example, patients with metastatic colorectal cancer undergoing bevacizumab treatment exhibited elevated plasma levels of PIGF, FGF and PDGF²²⁹ (Fig. 1c). Among the immunosuppressive signalling cues, PI3K γ and signal transducer and activator of transcription 3 (STAT3) and STAT6 activation in myeloid cells have been shown to render them non-responsive to VEGF blockade^{230–232}. Neutrophil extracellular traps have been shown to increase microvessel density through the neutrophil extracellular trap-DNA receptor CCDC25 on ECs, although their inhibition significantly inhibited tumour growth in a mouse model of gastric cancer²³³. In addition, the immune checkpoint, PD1 ligand 1 (PDL1), a negative regulator of antitumour immunity, was induced in immune cells²³⁴ as well as in tumour cells and stromal cells in tumours relapsing following anti-VEGF and anti-ANG2 therapy, eliciting T cell anergy or exhaustion, which was further exacerbated by T_{reg} cell infiltration^{235,236}.

Angiogenic ECs can also engender immunosuppressive features by eliciting barrier functions to reduce efficient lymphocyte infiltration²³⁷ (Fig. 6a). Thereby, TECs become anergic, do not respond to pro-inflammatory cytokines such as IL-1, interferon γ (IFN γ) and TNF, and downregulate lymphocyte and leukocyte adhesion molecules such as intercellular adhesion molecule 1, VCAM1, E-selectin and P-selectin, while upregulating galectins (for example, galectin 9), which are β -galactoside-binding proteins that are implicated in modulating cell–cell and cell–matrix interactions²²⁸. Such changes in the TEC surface profile can be further accompanied by elevated levels of T cell immunoglobulin mucin receptor 3 (ref. 238), PDL1 (refs. 239,240) and FAS ligand (FASL)²⁴¹, which have the capacity to either inhibit or kill EC-binding lymphocytes. TECs also upregulate MYC target protein 1 (MYCT1), through the transcriptional activator ETS translocation variant 2; MYCT1 encodes a transmembrane protein, which regulates actin cytoskeleton dynamics and promotes EC migration, hindering lymphocyte infiltration²⁴². Overall, tumour vessels diminish their adhesive surface for lymphocytes and suppress their activity.

A ground-breaking revelation was the discovery that VEGF and other angiogenic factors such as FGF and ANG2, besides their vessel-growth-promoting properties, can elicit direct immunosuppressive functions^{213,243}. For example, VEGF via VEGFR2 inhibits dendritic cell maturation²⁴⁴ and suppresses proliferation and cytotoxic function of effector T cells²⁴⁵ and also induces the proliferation of T_{reg} cells^{246,247} and MDSCs²⁴⁸. Moreover, VEGF and FGF2 downregulate the TNF-induced leukocyte adhesion molecules intercellular adhesion molecule 1 and VCAM1 on ECs, and VEGF can inhibit endothelial expression of lymphocyte-attracting cytokines (CXCL9, CXCL10, CXCL11 and CCL5), thereby reducing leukocyte infiltration^{249,250}.

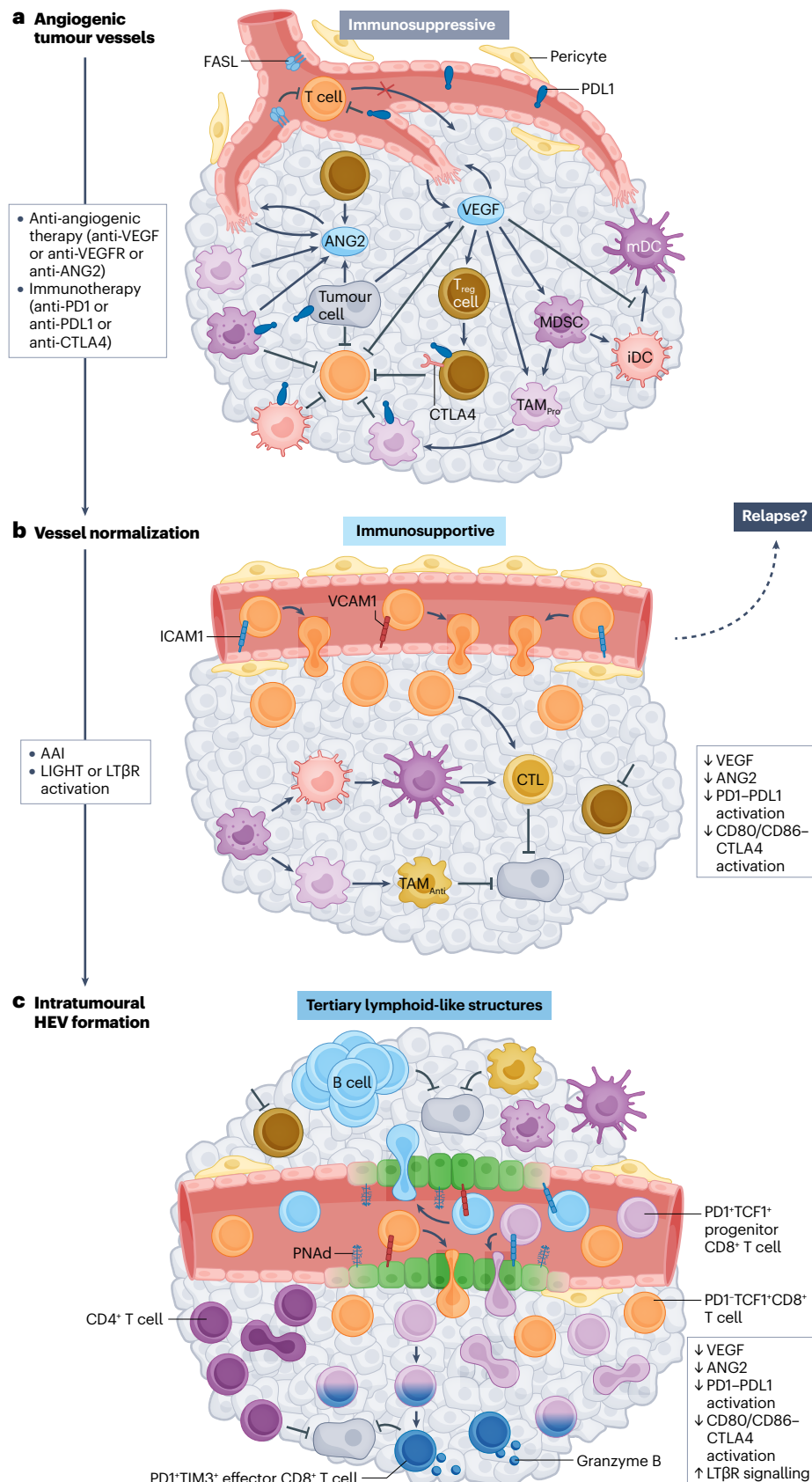


Fig. 6 | Remodelling vessels for efficient antitumour immunity. Vascular-immune interactions are crucial for promoting tumour growth advantages. **a**, Tumour blood vessels express FAS ligand (FASL) and programmed cell death protein 1 ligand 1 (PDL1), which inhibit lymphocyte infiltration. Sprouting endothelial cells (ECs) secrete high levels of vascular endothelial growth factor (VEGF), further increasing angiogenesis and immune suppression, via the recruitment and differentiation of myeloid-derived suppressor cells (MDSCs) into activated pro-tumour-associated macrophages (TAM_{pro}) and immature dendritic cells (iDCs), the selective inhibition of DC maturation, the activation of regulatory T (T_{reg}) cells and the inhibition of effector T cell activity. Subsequently, effector T cell antitumour cytotoxic activity is blocked through activation of the programmed cell death protein 1 (PD1)-PDL1 axis by MDSCs, TAM_{pro}, T_{reg} cells, iDCs and tumour cells and of the CD80 and CD86-cytotoxic T lymphocyte-associated antigen 4 (CTLA4) axis by T_{reg} cells. In turn, tumour cells, MDSCs, TAM_{pro} and T_{reg} cells promote vascular remodelling through the secretion of angiopoietin-2 (ANG2) and VEGF, leading to angiogenesis in a feedforward cycle. **b**, Combining anti-angiogenic agents with immunotherapy, the so-called anti-angiogenic immunotherapy (AAI) targeting VEGF-VEGF receptor 2 (VEGFR2) and ANG2 together with anti-PD1-PDL1 and anti-CTLA4, can induce vessel normalization with an immunosupportive environment. Normalization allows the infiltration of lymphocytes by restoring surface expression of intercellular adhesion molecule 1 (ICAM1) and vascular cell adhesion molecule 1 (VCAM1) on tumour ECs. Low levels of VEGF and blockade of PD1-PDL1 and CTLA4 signalling restore DC maturation and TAM antitumour activity (TAM_{anti}), while blocking immunosuppression by T_{reg} cells. This leads to increased antitumour activity of cytotoxic lymphocytes (CTLs) and reduction of tumour growth. Although proven efficient in preclinical studies, relapse is still observed upon AAI in clinical trials, thus suggesting that resistance mechanisms occur. **c**, The formation and presence of intratumoural high endothelial venules (HEVs) (TU-HEVs) are beneficial to antitumour immunity and are associated with the presence of tertiary lymphoid-like structures (TLS-like) composed of B cells, CD4⁺ and CD8⁺ T cells and innate immune cells in several malignancies. Upon AAI and activation of the lymphotoxin β receptor (LTBR) via binding to an agonist or a member of the tumour necrosis factor cytokine family, LIGHT, tumour ECs are reprogrammed into TU-HEVs that mediate an active influx of lymphocytes within tumours and foster permissive lymphocyte niches for PD1⁺ and PD1⁺ T cell-specific transcription factor 1 (TCF1)⁺ CD8⁺ T cell progenitors that differentiate into granzyme B⁺ PD1⁺ T cell immunoglobulin and mucin-domain containing-3 (TIM3)⁺ CD8⁺ T effector cells. mDC, mature dendritic cell; PNAd, peripheral node addressin.

Anti-angiogenic immunotherapies

On the basis of the observations described earlier, it is perhaps not surprising that vessel-normalizing and in part also anti-angiocrine and potentially vessel-promoting therapies can have counteractive effects on intratumoural immunosuppressive cells and thereby support an immune-stimulating milieu^{243,251}. For example, by blocking VEGF signalling alone or in combination with anti-ANG2, TECs upregulated lymphocyte adhesion molecules enabling increased lymphocyte influx, accompanied by a more prevalent presence of mature dendritic cells and activated effector T cells, whereas T_{reg} cells and MDSCs were reduced in breast and pancreatic neuroendocrine tumours^{235,236,252} (Fig. 6b). ANG2 inhibition alone also increased lymphocyte influx, enhanced response to anti-PD1 therapy in mouse melanomas and rendered non-responsive tumours susceptible to immunotherapy^{253,254}. In line with the reciprocal relationship between the endothelial and immune cell compartments, immune-stimulating cells can likewise promote vessel-normalizing effects. Indeed, eosinophils contribute to tumour immunity by secreting chemoattractants for CD8⁺ T cells and generating an immune-stimulating environment. Transfer of activated eosinophils with T cells to mice was associated with vessel normalization in melanomas²⁵⁵. Consistent with these data, increased plasma eosinophil levels in patients with glioblastoma undergoing bevacizumab treatment was associated with improved OS²⁵⁶. T cells also affect vascular functionality, as recently shown²⁵⁷. Depletion of effector CD4⁺ T cells increased the permeability of tumour vessels concomitant with decreased pericyte coverage, whereas CD4⁺ T cell infusion into mice bearing tumour promoted vessel normalization by increasing pericyte coverage²⁵⁷. Thus, effector CD4⁺ T cells can sustain vascular vessel maturation and functionality in tumours.

Therefore, it is conceivable that combining anti-angiogenic agents with immunotherapy (which we call anti-angiogenic immunotherapy) should be synergistic in that vascular-normalizing effects by anti-angiogenic agents should increase lymphocyte infiltration and activation, whereas immunotherapy should sensitize tumours to and hinder evasion from anti-angiogenic therapy. Several anti-angiogenic-immunotherapeutic strategies in mouse tumour models have supported this concept. The combination of anti-VEGF or the dual anti-ANG2 and anti-VEGF antibody AV2 with anti-PDL1 (refs. 235,236) therapy had enduring effects compared with single treatments, accompanied by prolonged vessel normalization and CD8⁺ T cell infiltration and activity (Fig. 6b). However, addition of a PD1 or PDL1 antibody to AV2 to mice bearing autochthonous lung tumours was not beneficial and even accelerated progression of a fraction of the tumours owing to infiltration by PD1⁺ T_{reg} cells. Dual tumour-associated macrophage targeting with a macrophage colony-stimulating factor 1 receptor inhibitor plus cisplatin reduced the numbers of T_{reg} cells, redirected the PD1 antibody to CD8⁺ T cells and improved the efficacy of anti-angiogenic immunotherapy, achieving regression of most tumours²⁵⁴. Similarly, targeting myeloid cell infiltration or activation with PI3Kγ inhibitors or Gr1-depleting antibodies in conjunction with angiogenic blockade was superior to single treatments and prolonged survival in multiple mouse tumour models^{230,232,258}. Blockade of the chemokine PK2, a target of granulocyte-colony-stimulating factor, inhibited the recruitment of MDSCs, which rendered tumours sensitive to anti-angiogenic and anti-PD1 therapy by enhancing the antitumour activity of cytotoxic T cells²⁵⁹. In addition, genetically engineered mice developing colorectal cancer became resistant to anti-VEGF therapy upon colitis-induced inflammation owing to infiltration of neutrophils and MDSCs. In this case, tumour progression and metastasis could be suppressed by blocking

granulocyte-colony-stimulating factor or downstream PK2. In line with this finding, patients with colorectal cancer had significantly higher plasma PK2 levels than healthy volunteers, and high plasma Bv8 levels were inversely correlated with OS²⁶⁰.

Sensitizing tumours to anti-angiogenic therapies and ensuring that their action is maintained can also be achieved by directly inducing or enhancing immunostimulating factors and pathways. For example, the stimulator of interferon genes (STING) signalling pathway is present in ECs and correlates with enhanced T cell infiltration into tumours and prolonged survival of mice bearing human colon and breast cancer tumours. Consistent with this, STING agonists (cGAMP or ADU-S100 (also known as ML RR-S2 CDA)) normalized the tumour vasculature in mouse Lewis lung carcinoma and PyMT breast tumours concomitant with increased expression of genes encoding type I and II interferon and vascular stabilizing proteins (for example, angiopoietin 1 (*Angpt1*), *Pdgfrb* and collagen alpha-4 (*Col4a*)). Furthermore, STING agonists combined with VEGFR2 blockade and/or ICB (anti-PD1 or anti-CTLA4) led to complete regression of immunotherapy-resistant tumours in mice²⁶¹. The order of drug administration should also be considered, with recent reports pointing towards improved efficacy of immunotherapy when delivered after angiogenic inhibitors²⁶².

The concept of vascular cooperation in improving the response rates and duration of ICB has been predominantly tested with bevacizumab and anti-PDL1 therapy (atezolizumab) in patients with cancer^{243,251,263}. Patients with renal cell carcinoma exhibited significantly enhanced PFS with the anti-angiogenic-ICB combination compared with standard sunitinib treatment^{264,265}. Importantly, these clinical studies also revealed that gene expression signatures associated with angiogenic activity, T-effector cells and IFNγ response and myeloid inflammation were strongly associated with PFS, whereas tumour mutation and neoantigen burden were not, confirming the results of various tumour model systems^{266–269}. The IMpower150 trial in patients with NSCLC revealed that the addition of atezolizumab to bevacizumab and chemotherapy (carboplatin plus paclitaxel) showed significant improvements in PFS and OS versus the standard-of-care bevacizumab plus chemotherapy by reducing risk of death by 22% and disease progression by 38% (refs. 270–272). Similar results were obtained in the IMbrave150 phase III trial in patients with hepatocellular carcinoma, in which atezolizumab plus bevacizumab demonstrated 42% reduction in risk of death and 41% reduction in progression when compared with the angiokinase inhibitor sorafenib^{273–275}. Most recently, this combination was also very efficacious in the adjuvant setting in the IMbrave050 trial, which was designed to delay or prevent recurrence in patients with high-risk hepatocellular carcinoma. Atezolizumab plus bevacizumab versus active surveillance after curative-intent resection or ablation of hepatocellular carcinoma yielded a 38% improvement in recurrence-free survival of patients^{276–278}.

Although these trials have yielded encouraging results with anti-angiogenic-ICB therapy, not all trials have demonstrated a favourable outcome. For example, ICB and anti-VEGF combinatorial treatment in patients with glioblastoma has been ineffective^{279,280}, as already predicted from a preclinical glioblastoma model system²³⁵. Various reasons may account for the differing responses, which likely include the level and type of immunosuppression and angiogenic activity occurring in different tumour types²⁸¹. Therefore, one can envision targeting other immune-regulating cues (for example, other immune checkpoint molecules, immunosuppressive metabolites and chemokines) depending on the tumour type. Similarly, as discussed earlier, there are

multiple possibilities to target the vasculature beyond VEGF inhibition. Such a well-designed and fine-tuned strategy, tailored to the tumour microenvironment of a particular tumour type, may likely generate the desired antitumour effects, but will also require a more thorough understanding of the versatile players in the intricate relationship of tumour vessels and the immune system.

Tumour high endothelial venule induction

Notably, anti-angiogenic immunotherapies and immunomodulating combination therapies with a strong antitumour immune response had additional vascular effects besides vessel normalization as they induce high endothelial venules (HEVs) in tumours (TU-HEVs)^{282–284}. TU-HEVs are reminiscent of organotypic HEVs in secondary lymphoid organs (which include lymph nodes, tonsils and Peyer's patches). They display peripheral node addressin, a specific modification found on several glycoproteins (sialomucins) decorated with 6-sulfo sialyl Lewis X-capped O-glycans to promote trafficking of L-selectin + lymphocytes into secondary lymphoid organs for their education and priming^{285–288}. As TU-HEVs also upregulate ligands for P-selectin and E-selectin, they can promote the extravasation of naive, memory and activated lymphocytes into the tumour and thereby support the formation of T and B cell lymphocyte aggregates that are reminiscent of tertiary lymphoid structures (TLSs)^{284,289–291} (Fig. 6c). TLSs can range from diffuse T and B cell aggregates to organized structures with T cells surrounding B cells that form primary follicles or secondary follicles with a germinal centre^{292–294}. Both TLSs and TU-HEVs have been commonly associated with a good prognosis in several tumour types including melanomas, NSCLC, gastric cancer and breast cancer^{284,290,292,295}. Thereby, TU-HEV density, concomitant with the magnitude of TLSs, exhibited prolonged metastasis-free, relapse-free or OS rates and has been predictive of response to ICB in soft-tissue sarcomas, renal cell carcinomas and melanomas^{296,297}. So far, there have only been two descriptions in which the presence of TU-HEVs and TLSs seems to correlate with a worse prognosis and this is in hepatocellular carcinoma²⁹⁸ and head and neck cancer²⁹⁹. However, these studies implied the presence of immature hyperdilated HEVs in lymph node metastases, suggestive of a phenotype in which HEVs lose their fate³⁰⁰.

Although there is no direct evidence that ectopic TU-HEV induction is required for TLS formation, the strong correlation between the presence of TU-HEVs and the extent of intratumoural B and T cell infiltration suggests that HEVs are key portals of lymphocyte entry enabling TLS formation, leading to an improved immune response. As a result, different strategies have been implemented in mouse tumours to promote the generation of TU-HEVs to turn 'cold tumours' into 'hot tumours'. Lymphotoxin $\alpha 1\beta 2$ (LT $\alpha 1\beta 2$) or LIGHT (also known as TNFSF14), produced by T cells, B cells and natural killer cells, appears to be the main driver inducing TU-HEVs by activating the endothelial LT β receptor (LT β R) and the downstream non-canonical nuclear factor- κ B pathway, which in turn induces the expression of genes involved in lymphoid organogenesis, such as those encoding HEV-specific enzymes for the glycocalyx and chemokines for lymphocyte attraction^{235,284,290,301–303}. To date, therapeutic TU-HEV induction has been effectively demonstrated by activating endothelial LT β R in conjunction with other immune-stimulating therapies such as anti-angiogenic agents or ICB²⁸². For example, several studies have demonstrated that the LT β R agonist, alone or in combination with anti-VEGFR2 plus anti-PDL1, or anti-CTLA4, or combined ICB, drives TU-HEV formation in association with an increased CD8⁺ T cell influx

Glossary

Anergy

In the context of T cells, it is a tolerance mechanism in which the lymphocyte is functionally inactivated following an antigen encounter. In the context of endothelial cells, it is a state in which endothelial cells no longer respond to inflammatory cytokines.

Angiocrine

Derived from endothelial cells.

Biosimilar

Non-generic versions of brand-name drugs that may offer more affordable treatment options to patients.

Metronomic chemotherapy

Continuous and dose-dense administration of chemotherapeutic drugs.

Mural cells

Pericytes and smooth muscle cells that surround and support blood vessels.

Pericrine

Derived from pericytes.

Pericytes

Mural cells covering blood microvessels, including capillaries.

Secondary lymphoid organs

Encapsulated structures in which lymphocytes mount adaptive immune responses.

Stalk cells

Endothelial cells that trail tip cells, proliferate and elongate the vascular sprout.

Tertiary lymphoid structures

Ectopic lymphoid tissues that drive antigen-specific immune responses at sites of chronic inflammation.

Tip cells

Migratory endothelial cells with long filopodia at the tip of the vascular sprout.

and an antitumour response in mouse breast, melanoma, glioblastoma, colon and sarcoma tumour models^{235,284,290,301,304}. An alternative approach of tumour endothelial LT β R activation via binding to a LIGHT peptide either as a single therapy or in combination with anti-PD1 led to HEV formation in mouse primary lung tumours, neuroendocrine pancreatic tumours, glioblastoma and lung metastases^{302,303}. In addition, immunostimulating therapies such as the STING agonist ADU-S100 (ref. 305), the poly (ADP-ribose) polymerase 1 (PARP) inhibitor BMN 673 (ref. 306) or the PI3K δ -inhibitor PI-3065 (ref. 307) also provoked TU-HEV induction concomitant with an antitumour immune response. Although the latter strategies do not directly activate LT β R, they endorse an influx of activated LT-positive lymphocytes that trigger EC conversion into HEVs. In support of this, the STING agonist ADU-S100 upregulated CXCL10, and the TLS-inducing factors CCL19, CCL21, LT α , LT β and LIGHT in B16 mouse melanoma³⁰⁵ and the PI3K δ -inhibitor PI-3065 targeted T_{reg} cells, which in turn enhanced CD8⁺ T cell activity and promoted HEV neogenesis³⁰⁷. T_{reg} cells may suppress TU-HEVs either directly or by creating an immunosuppressive environment. For example, immunosuppressive neutrophils also restrained TU-HEV formation in a KRAS-G12D-driven lung cancer mouse model³⁰⁸. In addition, a recent study demonstrated that loss of the genomic organizer special AT-rich sequence-binding protein 1 in CD4⁺ T cells promotes inducible T cell co-stimulator expression and thereby induces LIGHT⁺CXCL13⁺IL-21⁺ follicular helper T cell differentiation, resulting in the generation of HEV-containing TLSs in ovarian tumours in mice³⁰⁹.

Besides enabling lymphocyte influx, TU-HEVs generate permissive perivascular niches for PD1^{neg} and PD1⁺T cell-specific transcription factor 1⁺CD8⁺ T cell progenitor exhausted cells that were shown

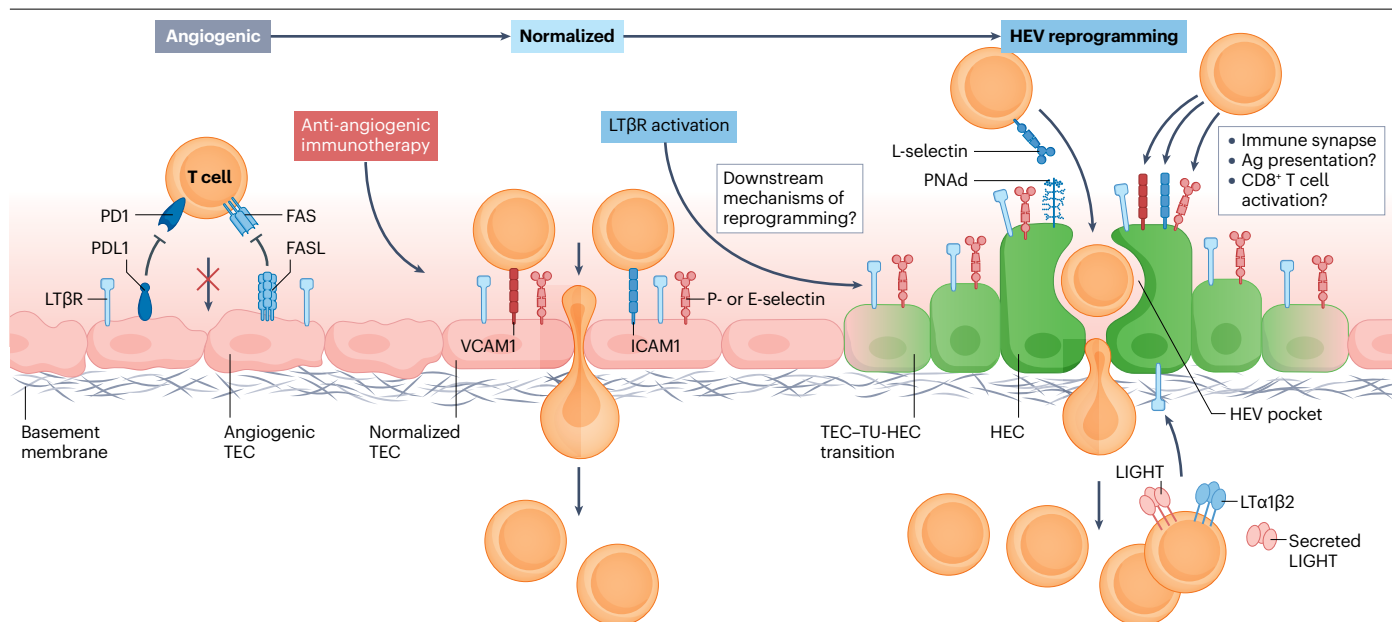


Fig. 7 | A simplistic model for reprogramming tumour vessels into high endothelial venules. Therapies such as anti-angiogenic therapy in combination with immune checkpoint blockade or other immunomodulating therapies stimulate the lymphotoxin β receptor (LT β R) axis in endothelial cells, normalizing tumour vessels and thereby enhancing recruitment of CD8⁺ T and natural killer cells. These cells further trigger endothelial LT β R activation in a feedforward loop via membrane-bound lymphotoxin α 1 β 2 (LT α 1 β 2) or LIGHT, which upon a certain threshold leads to the transitioning of tumour endothelial cells (TECs) into tumour-associated high endothelial venules (TU-HEVs; see Fig. 6 for the specifics of the immune landscape associated with TU-HEVs). Although several therapeutic strategies are currently used to induce TU-HEVs²⁸²

(Fig. 6), further studies are needed to decipher the molecular mechanisms of the LT β R-driven fate change. The expression of HEV-specific sialomucins such as peripheral node addressin (PNAd), P-selectin and E-selectin are putative crucial regulators of lymphocyte homing into the tumour. TU-HEVs may exert additional immunomodulating functions during lymphocyte transmigration within HEV pockets, whereby TU-HEVs could potentially present antigens, thereby supporting CD8⁺ T cell activation or tolerance. Ag, antigen; FASL, FAS ligand; HEC, high endothelial cell; ICAM1, intercellular adhesion molecule 1; PD1, programmed cell death protein 1; PDL1, PDL1 ligand 1; TU-HEC, tumour-associated high endothelial cell; VCAM1, vascular cell adhesion molecule 1.

to proliferate and differentiate into granzyme B⁺PD1⁺CD8⁺ T effector exhausted cells in response to anti-angiogenic immunotherapy or ICB combinations, sustaining the beneficial effects of ICB^{284,290,310,311}. TU-HEVs can also interact with other components of the tumour micro-environment. A recent study revealed that cutaneous sensory nerves in melanoma disrupted the formation of TU-HEV and mature TLSs as denervation partly relieved the inhibition and led to an increased influx and activation of T-effector cells, which was suppressed upon CD8⁺ T cell depletion³¹².

On the basis of the various observations described earlier, it has been proposed that not only HEVs regulate lymphocyte extravasation but also lymphocytes regulate HEV induction and maintenance in tumours. Therefore, sufficient LT-presenting CD8⁺ T lymphocytes and natural killer cells may have to assemble to reach the necessary threshold that enables endothelial conversion into HEVs. Further signalling cues between immune cells and ECs and fibroblasts are then required to form overt mature TLSs (Figs. 6 and 7).

Conclusions and future perspective

The diverse and versatile strategies of therapeutically targeting the tumour vasculature highlight the pivotal and multifaceted functions of the endothelium in influencing tumour propagation and progression and make it a pivotal component in the lifespan of cancer. Our perspective of how to target the tumour vasculature has changed substantially

over the last decades, not only because of the increased appreciation of its complexity but also because of the recognition of intricate and reciprocal interactions with other cell types, including the immune cell compartment. Independent of the treatment strategy, inducing tumour reduction by restoring tumour vessel functionality has predominantly improved standard of care and immunotherapy in several cancer types. Nevertheless, obtaining and identifying a suitable therapeutic window with the appropriate drug concentration remains challenging. No less relevant is the fact that tumour cells can grow along blood vessels and can adapt to do so if needed. However, tumour co-opted blood vessels behave like normal quiescent blood vessels and, as such, are very challenging to treat. To date, strategies to specifically target co-opted blood vessels have been unsuccessful, but the future development of these will be of high importance for the treatment of tumours and metastatic lesions that intrinsically or in response to therapies co-opt blood vessels to grow and disseminate. A promising strategy may lie in targeting angiocrine and/or juxtacrine communication signals with tumour cells that could lead to a selective disruption of the co-opted and angiogenic tumour vascular niche without compromising the integrity of the pre-existing vasculature or even promote vessel growth to overcome hypoxia-dependent resistance and aggressive tumour regrowth. In addition, it will be important for future studies to consider the perivascular cell constituents, the pericytes and smooth muscle cells, which interact not only with ECs but also with immune

cells and other cell constituents of the tumour microenvironment as well as tumour cells, partly by producing pericrine factors³¹³. Although many studies have identified novel communication cues in preclinical cancer models, the applicability of targeting such factors in the clinic still remains an open question. The encouraging preclinical results of therapeutically transitioning TECs into TU-HEVs and generating TLSs, which can turn a cold tumour into a hot tumour, also provide a strong rationale for deepening our understanding of the immune cell–EC axis in eliciting and improving anticancer immune responses³¹⁴. One can envision additional implications of TECs in regulating immunity^{4,315,316}. For example, although ECs are not professional antigen-presenting cells, they possess the capacity to express major histocompatibility complex class I and II molecules, specifically upon inflammation, and can thereby act like antigen-presenting cells to activate T cells or promote tolerance^{317–320} (Fig. 7). Moreover, blood vessels, being a central and pivotal compartment in tumours, interact with various other cell types besides immune cells. One example, which intimately interacts with the vasculature and shares similar signals for development, growth and differentiation, is the nervous system³²¹. This is most prominently exemplified by the different VEGF ligands that promote both vascular and neuronal growth³²². The innervation of tumours has been recently proposed as an additional hallmark-inducing constituent of the tumour microenvironment because it has multifaceted effects^{19,323}. Adrenergic nerves of the sympathetic nervous system in prostate cancer, pancreatic ductal adenocarcinoma or glioblastoma not only appear to promote proliferation of cancer cells but can also promote tumour inflammation and support tumour angiogenesis^{324–326}. These nerves are in close juxtaposition with blood vessels because they grow together during development whereby adrenergic nerves guide the vascular structure of organs³²⁷. Given this crosstalk, it is conceivable that intratumoural nerves and vessels will affect each other in response to anti-angiogenic, nerve-targeting or immunomodulating therapies and so this warrants further investigation. Taken together, the studies described in this Review are testimony to the pivotal and complex roles of vascular cells in tumorigenesis and how different vascular treatment strategies can affect different tumour microenvironment constituents and subsequent tumour behaviour. The emerging trend of using single and spatial omics analyses of tumours in the context of treatment strategies targeting the tumour microenvironment will likely provide a more holistic view and new understanding of the intricate relationship between the tumour vasculature and other constituents of the tumour microenvironment.

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Author contributions

The authors contributed equally to all aspects of the article.

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K.H.-D. is a consultant for Ellipesis, RGD-Science and Vasodynamics. G.B. is a consultant for Mestag. S.G. declares no competing interests.

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